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CLOUD COMPUTING

Lab 5

Task 1

Discover missing command & install Java using apt suggestion

```
alishba@ubuntu:~$ ssh alishba@192.168.145.131
PS C:\Users\zaina> ssh alishba@192.168.145.131
>>
alishba@192.168.145.131's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 24 Oct 06:53:06 UTC 2025

System load:  0.01               Processes:    212
Usage of /:   15.7% of 29.36GB   Users logged in: 1
Memory usage: 13%              IPv4 address for ens33: 192.168.145.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sun Sep 28 15:15:23 2025 from 192.168.145.1
alishba@ubuntu:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.14+7-1~24.04, or
sudo apt install openjdk-21-jre-headless # version 21.0.6+7-1~24.04.1
sudo apt install default-jre             # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.26+4-1ubuntu1~24.04
sudo apt install openjdk-8-jre-headless  # version 8u442-b06~us1-0ubuntu1~24.04
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
alishba@ubuntu:~$

Adding debian:Starfield_Root_Certificate_Authority_-_G2.pem
Adding debian:Starfield_Services_Root_Certificate_Authority_-_G2.pem
Adding debian:SwissSign_Gold_CA_-_G2.pem
Adding debian:SwissSign_Silver_CA_-_G2.pem
Adding debian:SZAFIR_Root_CA2.pem
Adding debian:Telia_Root_CA_v2.pem
Adding debian:TeliaSonera_Root_CA_v1.pem
Adding debian:TrustAsia_Global_Root_CA_G3.pem
Adding debian:TrustAsia_Global_Root_CA_G4.pem
Adding debian:Trustwave_Global_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P256_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P384_Certification_Authority.pem
Adding debian:T-TeleSec_GlobalRoot_Class_2.pem
Adding debian:T-TeleSec_GlobalRoot_Class_3.pem
Adding debian:TUBITAK_Kamu_SM_SSL_Kok_Sertifikasi_-_Surum_1.pem
Adding debian:TunTrust_Root_CA.pem
Adding debian:TWCA_Global_Root_CA.pem
Adding debian:TWCA_Root_Certification_Authority.pem
Adding debian:UCA_Extended_Validation_Root.pem
Adding debian:UCA_Global_G2_Root.pem
Adding debian:USERTrust_ECC_Certification_Authority.pem
Adding debian:USERTrust_RSA_Certification_Authority.pem
Adding debian:vTrus_ECC_Root_CA.pem
Adding debian:vTrus_Root_CA.pem
Adding debian:XRamp_Global_CA_Root.pem
done.
Setting up openjdk-21-jre:amd64 (21.0.8+9-us1-0ubuntu1~24.04.1) ...
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$
```

```

no VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
alishba@ubuntu:~$

```

```

libgdk-pixbuf2.0-common libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-3
libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpc
libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwayland-serv
libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0 lib
libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxrender1 libxshmfence1 libxt6t64 libxtst6 libxv1 libxx
session-migration ubuntu-mono x11-common x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
default-jre-headless* openjdk-21-jre-headless*
0 upgraded, 0 newly installed, 2 to remove and 67 not upgraded.
After this operation, 203 MB disk space will be freed.
Reading database ... 101544 files and directories currently installed.)
Removing default-jre-headless (2:1.21-75+exp1) ...
Removing openjdk-21-jre-headless:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...
Reading database ... 101253 files and directories currently installed.)
Purging configuration files for openjdk-21-jre-headless:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...
alishba@ubuntu:~$

```

```

alishba@ubuntu:~$ hash -r
alishba@ubuntu:~$ java
Command 'java' not found, but can be installed with:
sudo apt install default-jre # version 2:1.17-75, or
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install openjdk-11-jre-headless # version 11.0.28+6~1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
alishba@ubuntu:~$

```

Task 2

Install & remove Java using apt-get (explicitly)

```

alishba@ubuntu:~$ hash -r
alishba@ubuntu:~$ java
Command 'java' not found, but can be installed with:
sudo apt install default-jre # version 2:1.17-75, or
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install openjdk-11-jre-headless # version 11.0.28+6~1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
alishba@ubuntu:~$

```

```

alishba@ubuntu:~$ hash -r
alishba@ubuntu:~$ java
command 'java' not found, but can be installed with:
sudo apt install default-jre # version 2:1.17-75, or
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
alishba@ubuntu:~$ s

```

Task 3

apt update vs apt upgrade - run & explain

```

alishba@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
67 packages can be upgraded. Run 'apt list --upgradable' to see them.
alishba@ubuntu:~$

```

```

alishba@ubuntu: ~
update-initramfs: Generating /boot/initrd.img-6.8.0-86-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file /etc/default/grub
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-86-generic
Found initrd image: /boot/initrd.img-6.8.0-86-generic
Found linux image: /boot/vmlinuz-6.8.0-71-generic
Found initrd image: /boot/initrd.img-6.8.0-71-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-71-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart cron.service multipathd.service polkit.service upower.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
alishba @ session #1: login[1315]
alishba @ session #4: apt[4727], sshd[1718,1800]
alishba @ user manager service: systemd[1397]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$

```

```
Select alishba@ubuntu: ~
Pending kernel upgrade!
Running kernel version:
6.8.0-71-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart cron.service multipathd.service polkit.service upower.service

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systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
alishba @ session #1: login[1315]
alishba @ session #4: apt[4727], sshd[1718,1800]
alishba @ user manager service: systemd[1397]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$ nano ~/apt_update_vs_upgrade.md
alishba@ubuntu:~$ cat ~/apt_update_vs_upgrade.md
nano ~/apt_update_vs_upgrade.md

The command 'apt update' updates the local package index on the system.
It downloads the latest information about available packages and versions from the repositories but does not install or upgrade anything.
The command 'apt upgrade' uses the updated package index to install the newest versions of all installed packages.
Without running 'apt update' first, 'apt upgrade' may use outdated package lists and miss new updates.

alishba@ubuntu:~$ cat ~/apt_update_vs_upgrade.md
nano ~/apt_update_vs_upgrade.md

The command 'apt update' updates the local package index on the system.
It downloads the latest information about available packages and versions from the repositories but does not install or upgrade anything.
The command 'apt upgrade' uses the updated package index to install the newest versions of all installed packages.
Without running 'apt update' first, 'apt upgrade' may use outdated package lists and miss new updates.

alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo snap install --classic code
[sudo] password for alishba:
2025-10-24T08:08:51Z INFO Waiting for automatic snapd restart...
code 7d842fb8 from Visual Studio Code (vscode✓) installed
alishba@ubuntu:~$
```

Task 4

Install Visual Studio Code via snap on CLI and verify (DO NOT remove Code)

```
alishba@ubuntu:~$ snap list code
Name Version Rev Tracking Publisher Notes
code 7d842fb8 211 latest/stable vscode✓ classic
alishba@ubuntu:~$
```

```
code 7d842fb8 211 latest/stable vscode✓ classic
alishba@ubuntu:~$ code --version
1.105.1
7d842fb85a0275a4a8e4d7e040d2625abbf7f084
x64
alishba@ubuntu:~$
```

```
alishba@ubuntu: ~  
alishba@ubuntu:~$ cat ~/apt_update_vs_upgrade.md  
nano ~/apt_update_vs_upgrade.md  
  
The command 'apt update' updates the local package index on the system.  
It downloads the latest information about available packages and versions from the repositories but does not install or upgrade anything.  
The command 'apt upgrade' uses the updated package index to install the newest versions of all installed packages.  
Without running 'apt update' first, 'apt upgrade' may use outdated package lists and miss new updates.  
  
alishba@ubuntu:~$ sudo snap install --classic code  
[sudo] password for alishba:  
2025-10-24T08:08:51Z INFO Waiting for automatic snapd restart...  
code 7d842fb8 from Visual Studio Code (vscode) installed  
alishba@ubuntu:~$ snap list code  
Name Version Rev Tracking Publisher Notes  
code 7d842fb8 211 latest/stable vscode/ classic  
alishba@ubuntu:~$ code --version  
1.105.1  
7d842fb85a0275a4a8e4d7e040d2625abbf7f084  
x64  
alishba@ubuntu:~$ snap info code  
name: code  
summary: Code editing. Redefined.  
publisher: Visual Studio Code (vscode/)  
store-url: https://snapcraft.io/code  
contact: https://twitter.com/code  
license: unset  
description: |  
Visual Studio Code is a new choice of tool that combines the  
simplicity of a code editor with what developers need for the core  
edit-build-debug cycle.  
commands:  
- code  
- code.url-handler  
snap-id: Ht0aUHi7ofh9Fbwh6m7jUN2pAy6kz8iu  
tracking: latest/stable  
refresh-date: today at 08:09 UTC  
channels:  
latest/stable: 7d842fb8 2025-10-15 (211) 346MB classic  
latest/candidate: ↑  
latest/beta: ↑  
latest/edge: ↑  
installed: 7d842fb8 (211) 346MB classic  
alishba@ubuntu:~$ S
```

```
alishba@ubuntu:~$ ls -l /snap/bin | grep code  
lrwxrwxrwx 1 root root 13 Oct 24 08:09 code -> /usr/bin/snap  
lrwxrwxrwx 1 root root 13 Oct 24 08:09 code.url-handler -> /usr/bin/snap  
alishba@ubuntu:~$
```

```
Select alishba@ubuntu: ~  
x64  
alishba@ubuntu:~$ snap info code  
name: code  
summary: Code editing. Redefined.  
publisher: Visual Studio Code (vscode/)  
store-url: https://snapcraft.io/code  
contact: https://twitter.com/code  
license: unset  
description: |  
Visual Studio Code is a new choice of tool that combines the  
simplicity of a code editor with what developers need for the core  
edit-build-debug cycle.  
commands:  
- code  
- code.url-handler  
snap-id: Ht0aUHi7ofh9Fbwh6m7jUN2pAy6kz8iu  
tracking: latest/stable  
refresh-date: today at 08:09 UTC  
channels:  
latest/stable: 7d842fb8 2025-10-15 (211) 346MB classic  
latest/candidate: ↑  
latest/beta: ↑  
latest/edge: ↑  
installed: 7d842fb8 (211) 346MB classic  
alishba@ubuntu:~$ ls -l /snap/bin | grep code  
lrwxrwxrwx 1 root root 13 Oct 24 08:09 code -> /usr/bin/snap  
lrwxrwxrwx 1 root root 13 Oct 24 08:09 code.url-handler -> /usr/bin/snap  
alishba@ubuntu:~$ sudo apt update  
[sudo] password for alishba:  
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]  
Fetched 200 kB in 2s (106 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
alishba@ubuntu:~$
```

Task 5

Install XFCE GUI + XRDP - minimal desktop and remote access (GUI) and launch VS Code

```
alishba@ubuntu: ~  
aspell-autobuildhash: processing: en [en_AU-wo_accents-only].  
aspell-autobuildhash: processing: en [en_CA-variant_0].  
aspell-autobuildhash: processing: en [en_CA-variant_1].  
aspell-autobuildhash: processing: en [en_CA-w_accents-only].  
aspell-autobuildhash: processing: en [en_CA-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ise-w_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ize-w_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-variant_0].  
aspell-autobuildhash: processing: en [en_GB-variant_1].  
aspell-autobuildhash: processing: en [en_US-w_accents-only].  
aspell-autobuildhash: processing: en [en_US-wo_accents-only].  
Processing triggers for sgml-base (1.31) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.0) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
6.8.0-71-generic  
Diagnostics:  
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.  
  
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.  
  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
alishba @ session #1: login[1315]  
alishba @ session #4: sshd[1718,1800]  
alishba @ user manager service: systemd[1397]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
alishba@ubuntu:~$
```

```
alishba@ubuntu: ~  
Setting up xorgxrdp (1:0.9.19-1) ...  
Setting up libfuse2t64:amd64 (2.9.9-8.1build1) ...  
Setting up xrdp (0.9.24-4) ...  
  
Generating 2048 bit rsa key...  
  
ssl_gen_key_xrdp1 ok  
  
saving to /etc/xrdp/rsakeys.ini  
  
Created symlink /etc/systemd/system/multi-user.target.wants/xrdp-sesman.service → /usr/lib/systemd/system/xrdp-sesman.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/xrdp.service → /usr/lib/systemd/system/xrdp.service.  
Setting up pipewire-module-xrdp (0.2-2) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
6.8.0-71-generic  
Diagnostics:  
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.  
  
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.  
  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
alishba @ session #1: login[1315]  
alishba @ session #4: sshd[1718,1800]  
alishba @ user manager service: systemd[1397]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo systemctl restart xrdp
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

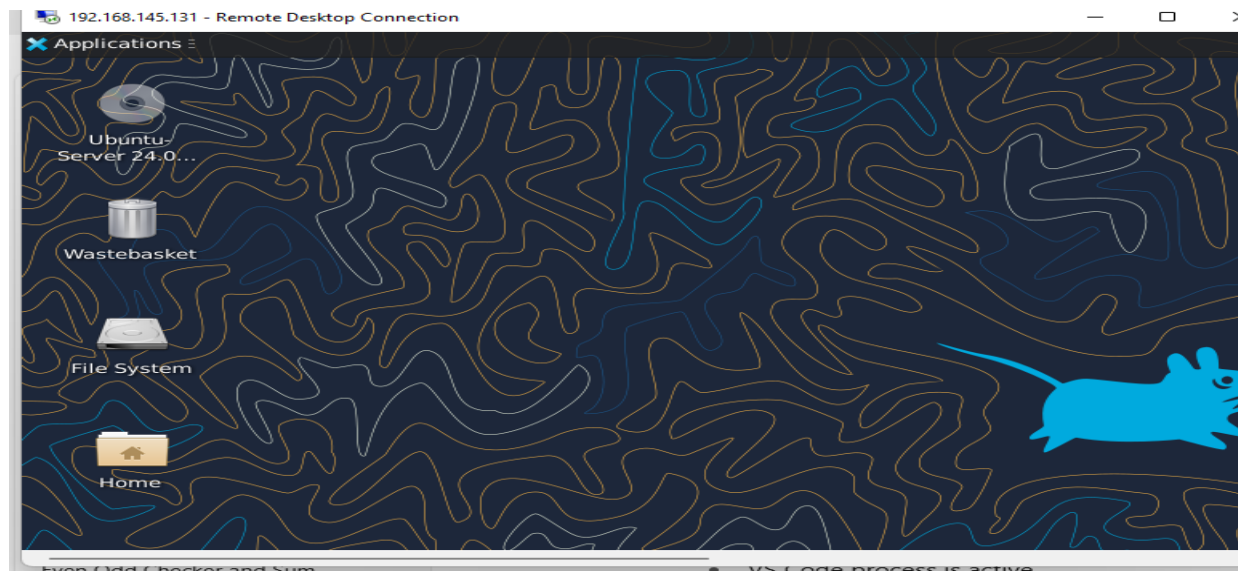
No containers need to be restarted.

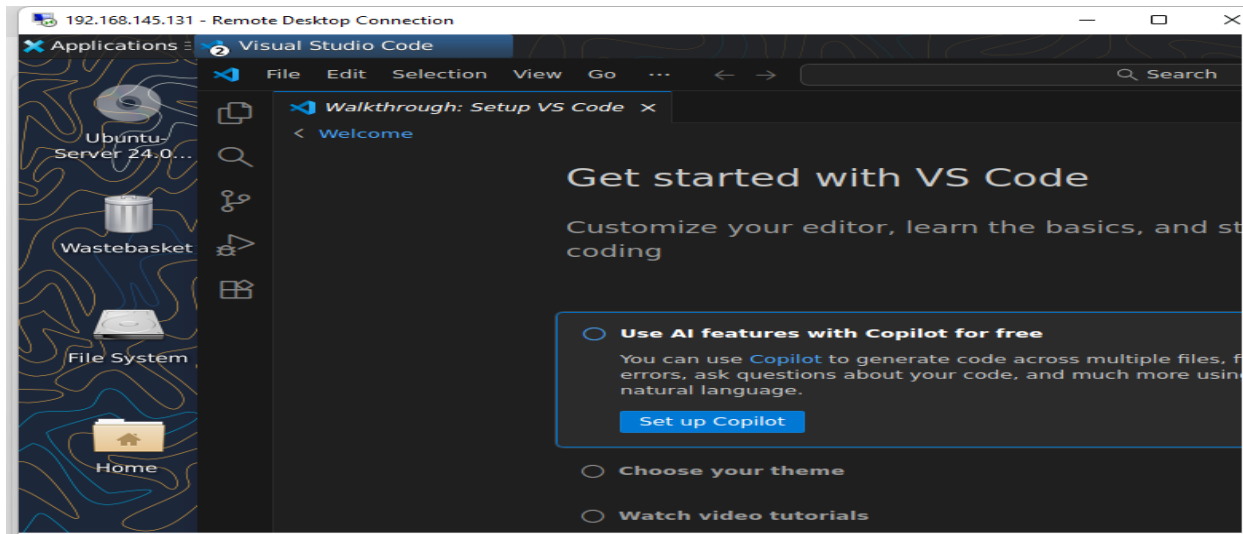
User sessions running outdated binaries:
alishba @ session #1: login[1315]
alishba @ session #4: sshd[1718,1800]
alishba @ user manager service: systemd[1397]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-10-24 08:38:41 UTC; 1min 49s ago
     Docs: man:xrdp(8)
           man:xrdp.ini(5)
   Process: 29087 ExecStartPre=/bin/sh /usr/share/xrdp/socksetup (code=exited, status=0/SUCCESS)
   Process: 29095 ExecStart=/usr/sbin/xrdp $XRDPOPTIONS (code=exited, status=0/SUCCESS)
  Main PID: 29097 (xrdp)
    Tasks: 1 (limit: 2210)
   Memory: 860.0K (peak: 1.5M)
      CPU: 24ms
   CGroup: /system.slice/xrdp.service
           └─29097 /usr/sbin/xrdp

Oct 24 08:38:40 ubuntu systemd[1]: Starting xrdp.service - xrdp daemon...
Oct 24 08:38:40 ubuntu xrdp[29095]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 08:38:40 ubuntu xrdp[29095]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 08:38:40 ubuntu xrdp[29095]: [INFO ] xrdp_listen_pp done
Oct 24 08:38:40 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such file or directory
Oct 24 08:38:41 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
Oct 24 08:38:42 ubuntu xrdp[29097]: [INFO ] starting xrdp with pid 29097
Oct 24 08:38:42 ubuntu xrdp[29097]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 08:38:42 ubuntu xrdp[29097]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 08:38:42 ubuntu xrdp[29097]: [INFO ] xrdp_listen_pp done
alishba@ubuntu:~$
```

```
Oct 24 08:38:42 ubuntu xrdp[29097]: [INFO ] xrdp_listen_pp done
alishba@ubuntu:~$ echo xfce4-session > ~/.xsession
alishba@ubuntu:~$
```





Task 6

Install lightdm-gtk-greeter and GUI verification - start GUI, open VS Code, take snapshot, then end (GUI)

```
alishba@ubuntu: ~  
Setting up gnome-themes-extra:amd64 (3.28-2ubuntu5) ...  
Processing triggers for hicolor-icon-theme (0.17-2) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
6.8.0-71-generic  
Diagnostics:  
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.  
  
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.  
  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/needrestart.d/dbus.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
alishba @ session #1: login[1315]  
alishba @ session #4: sshd[1718,1800]  
alishba @ user manager service: systemd[1397]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
alishba@ubuntu:~$
```



```
o VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu:~$ sudo mkdir -p /etc/lightdm/lightdm.conf.d
alishba@ubuntu:~$ echo "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user-timeout=0" | sudo tee /etc/lightdm/lightdm.conf.d/99-xfce.conf
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo rm -f /var/lib/lightdm/.Xauthority
-f ~/.Xauthority
sudo rm -rf ~/.cache/sessions
sudo chown -R $USER:$USER /home/$USER
sudo rm -f ~/.Xauthority
sudo rm -rf ~/.cache/sessions
sudo chown -R $USER:$USER /home/$USER
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo systemctl restart lightdm
alishba@ubuntu:~$
```

```
instance name specified.
alishba@ubuntu:~$ sudo systemctl enable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.
```

Possible reasons for having these kinds of units are:

- A unit may be statically enabled by being symlinked from another unit's .wants/, .requires/, or .upholds/ directory.
- A unit's purpose may be to act as a helper for some other unit which has a requirement dependency on it.
- A unit may be started when needed via activation (socket, path, timer, D-Bus, udev, scripted systemctl call, ...).
- In case of template units, the unit is meant to be enabled with some instance name specified.

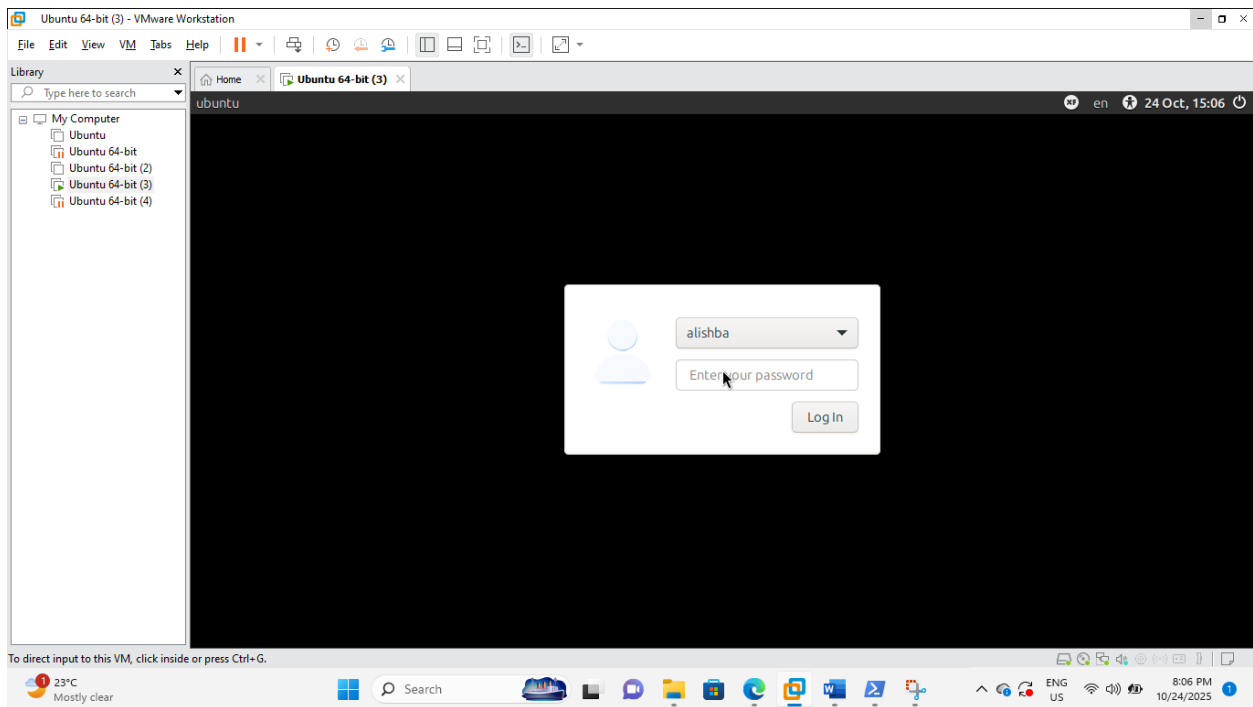
```
alishba@ubuntu:~$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo reboot
[sudo] password for alishba:
```

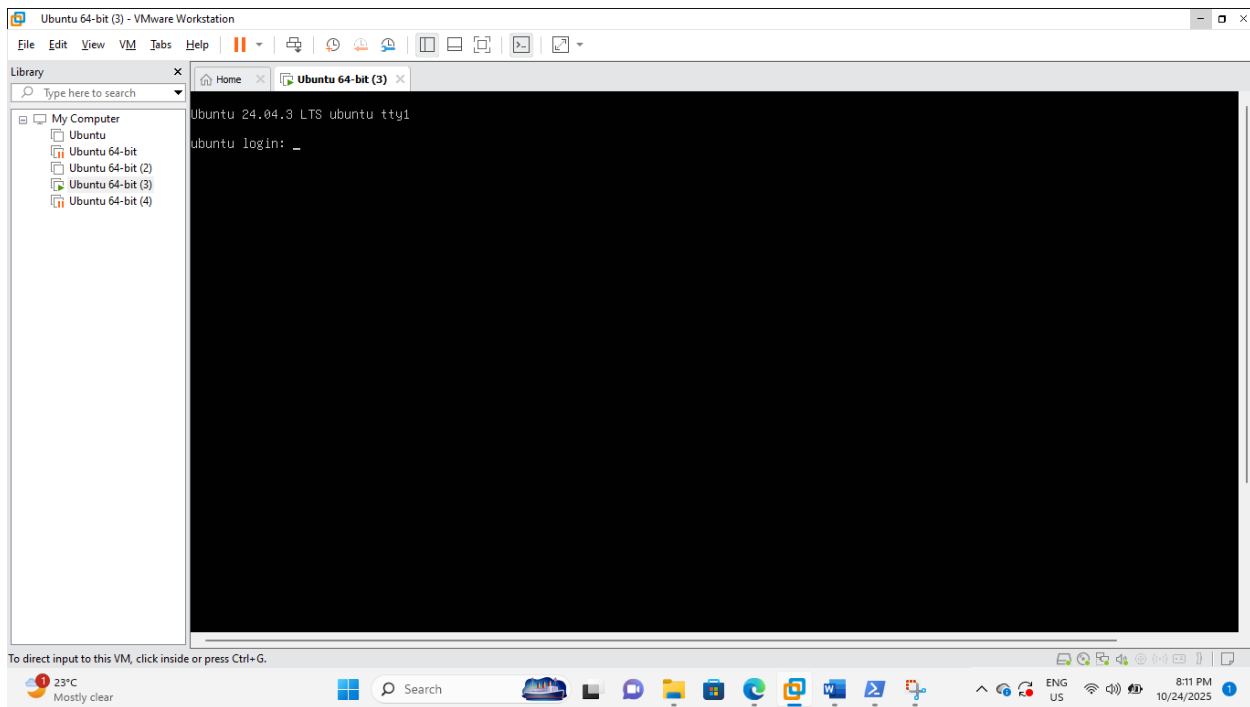
Broadcast message from root@ubuntu on pts/1 (Fri 2025-10-24 14:59:32 UTC):

The system will reboot now!

```
alishba@ubuntu:~$ Connection to 192.168.145.131 closed by remote host.
Connection to 192.168.145.131 closed.
PS C:\Users\zaina>
```

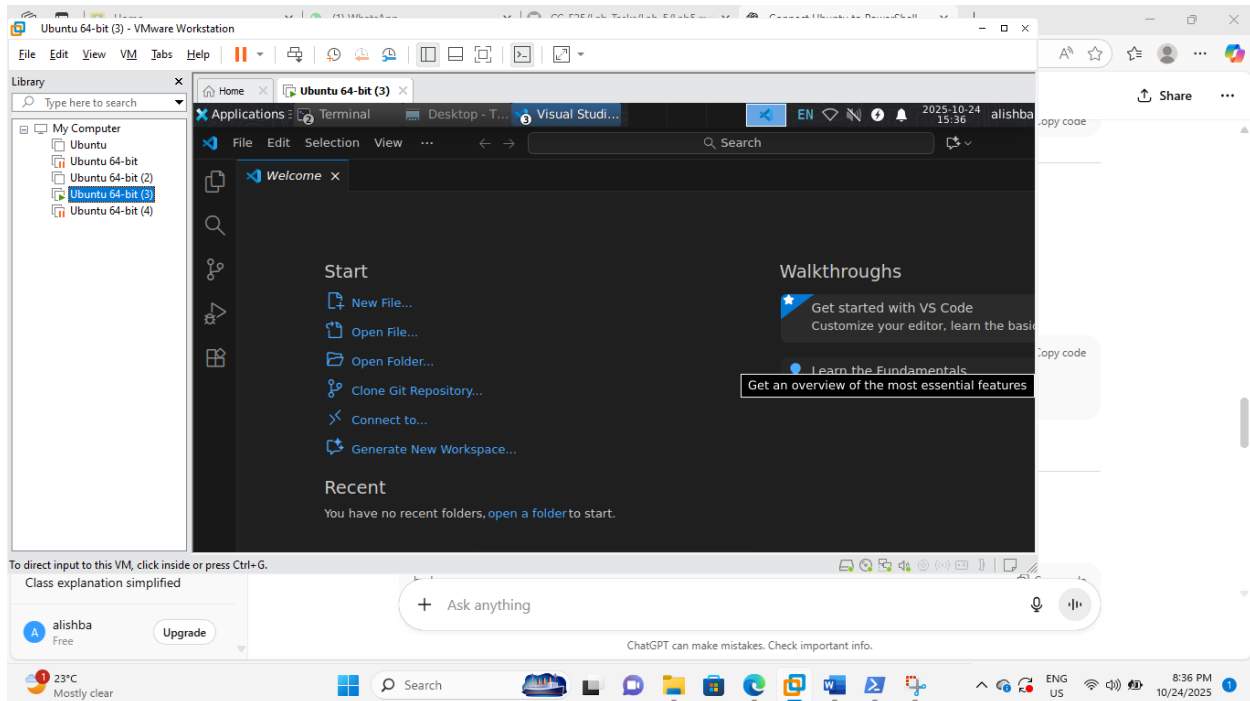


```
Last login: Fri Oct 24 15:05:20 2025 from 192.168.143.1
alishba@ubuntu:~$ sudo systemctl set-default multi-user.target
[sudo] password for alishba:
Sorry, try again.
[sudo] password for alishba:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
alishba@ubuntu:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
Removed "/etc/systemd/system/display-manager.service".
alishba@ubuntu:~$
```



```
Last login: Fri Oct 24 15:08:59 2025 from 192.168.145.1
alishba@ubuntu:~$ sudo systemctl start lightdm
[sudo] password for alishba:
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo systemctl stop lightdm
alishba@ubuntu:~$
```



Task 7

Install Google Chrome by adding its apt source & key (Chrome)

```
alishba@ubuntu:~$ sudo apt install google-chrome-stable -y
[sudo] password for alishba:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ ls -la /etc/apt
total 48
drwxr-xr-x  9 root root  4096 Sep 28 14:37 .
drwxr-xr-x 141 root root 12288 Oct 24 10:35 ..
drwxr-xr-x  2 root root  4096 Oct 24 07:55 apt.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 auth.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 keyrings
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d.save
-rw-r--r--  1 root root    70 Sep 28 14:39 sources.list
drwxr-xr-x  2 root root  4096 Sep 28 14:39 sources.list.d
drwxr-xr-x  2 root root  4096 Aug  5 17:01 trusted.gpg.d
alishba@ubuntu:~$ s
```

```
alishba@ubuntu:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
alishba@ubuntu:~$
```

```
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
alishba@ubuntu:~$ ls -la /etc/apt/sources.list.d/
total 16
drwxr-xr-x 2 root root 4096 Sep 28 14:39 .
drwxr-xr-x 9 root root 4096 Sep 28 14:37 ..
-rw-r--r-- 1 root root 382 Sep 28 14:39 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
alishba@ubuntu:~$ s_
```

```
alishba@ubuntu:~$ alishba@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg
```

```
alishba@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo apt update
google-chrome-stable -y
sudo apt install google-chrome-stable -y
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Err:2 http://dl.google.com/linux/chrome/deb stable InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 32EE5355A6BC6E42
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,536 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:10 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,164 B]
Get:11 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Reading package lists... Done
W: GPG error: http://dl.google.com/linux/chrome/deb stable InRelease: The following signatures couldn't be verified because the public key is not available: 32EE5355A6BC6E42
E: The repository 'http://dl.google.com/linux/chrome/deb stable InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
```

```
alishba@ubuntu:/tmp$ sudo dpkg -i /tmp/google-chrome-stable_current_amd64.deb
Selecting previously unselected package google-chrome-stable.
Reading database ... 179892 files and directories currently installed.)
Preparing to unpack .../google-chrome-stable_current_amd64.deb ...
Unpacking google-chrome-stable (141.0.7390.122-1) ...
Setting up google-chrome-stable (141.0.7390.122-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-www-browser (x-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gnome-www-browser (gnome-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/google-chrome (google-chrome) in auto mode
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
alishba@ubuntu:/tmp$ google-chrome --version
Google Chrome 141.0.7390.122
alishba@ubuntu:/tmp$
```

Task 8

Install applications via PPA (Audacity & OBS) and launch

```
alishba@ubuntu:~$ sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu/
Suites: noble
Components: main

Description:
Unofficial build of Audacity audio editor

For help, please use Audacity forum: http://forum.audacityteam.org/

If the packages here are helpful, you may buy me a coffee:

https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Get:5 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,064 B]
Get:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492 B]
Fetched 19.6 kB in 5s (3,851 B/s)
Reading package lists... Done
```

```
alishba@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ audacity
Command 'audacity' not found, but can be installed with:
sudo snap install audacity # version 3.7.1, or
sudo apt install audacity # version 2.4.2~dfsg0-5
See 'snap info audacity' for additional versions.
alishba@ubuntu:~$ sudo snap install audacity # version 3.7.1,
error: cannot perform the following tasks:
- Download snap "audacity" (1208) from channel "stable" (download too slow: 0.00 bytes/sec)
- Download snap "gnome-46-2404" (125) from channel "stable" (download too slow: 0.00 bytes/sec)
- Download snap "gtk-common-themes" (1535) from channel "stable" (download too slow: 0.00 bytes/sec)
- Download snap "mesa-2404" (1110) from channel "stable" (download too slow: 0.00 bytes/sec)
alishba@ubuntu:~$ sss
```

```
alishba@ubuntu:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
[sudo] password for alishba:
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Get:4 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:5 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 3s (7,458 B/s)
Reading package lists... Done
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
[sudo] password for alishba:
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Get:4 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:5 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 3s (7,458 B/s)
Reading package lists... Done
alishba@ubuntu:~$
```

```
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.454: file ../../gio/gfileinfo.c: line 1845 (g_file_info_get_size): should not be reached
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.454: GFileInfo created without standard::size
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: file ../../gio/gfileinfo.c: line 1845 (g_file_info_get_size): should not be reached
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: GFileInfo created without standard::size
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: file ../../gio/gfileinfo.c: line 1845 (g_file_info_get_size): should not be reached
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: GFileInfo created without standard::size
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: file ../../gio/gfileinfo.c: line 1845 (g_file_info_get_size): should not be reached
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: GFileInfo created without standard::size
(flatpak install:44676): Glib-GIO-CRITICAL **: 03:51:27.455: GFileInfo created without standard::size
```

ID	Branch	Op	Remote	Download
1. [✓] com.obsproject.Studio.Locale	stable	i	flathub	2.1 kB / 47.0 kB
2. [✓] org.freedesktop.Platform.GL.default	24.08	i	flathub	144.7 MB / 145.4 MB
3. [✓] org.freedesktop.Platform.GL.default	24.08extra	i	flathub	24.0 MB / 145.4 MB
4. [✓] org.freedesktop.Platform.openh264	2.5.1	i	flathub	911.0 kB / 971.4 kB
5. [✓] org.kde.KStyle.Adwaita	6.8	i	flathub	8.2 MB / 8.2 MB
6. [✓] org.kde.Platform.Locale	6.8	i	flathub	18.6 kB / 409.2 MB
7. [✓] org.kde.Platform	6.8	i	flathub	418.5 MB / 391.4 MB

Task 9

Create a Kubernetes sample YAML using vim


```
alishba@ubuntu:~/Lab5$ ls -la
total 12
drwxrwxr-x  2 alishba alishba 4096 Oct 25 04:07 .
drwxr-x--- 23 alishba alishba 4096 Oct 25 04:07 ..
-rw-rw-r--  1 alishba alishba  285 Oct 25 04:07 k8s-sample.yaml
alishba@ubuntu:~/Lab5$
```

Task 10

Edit the Kubernetes YAML - add annotation, verify, then discard temporary change

```

lishba@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always

```

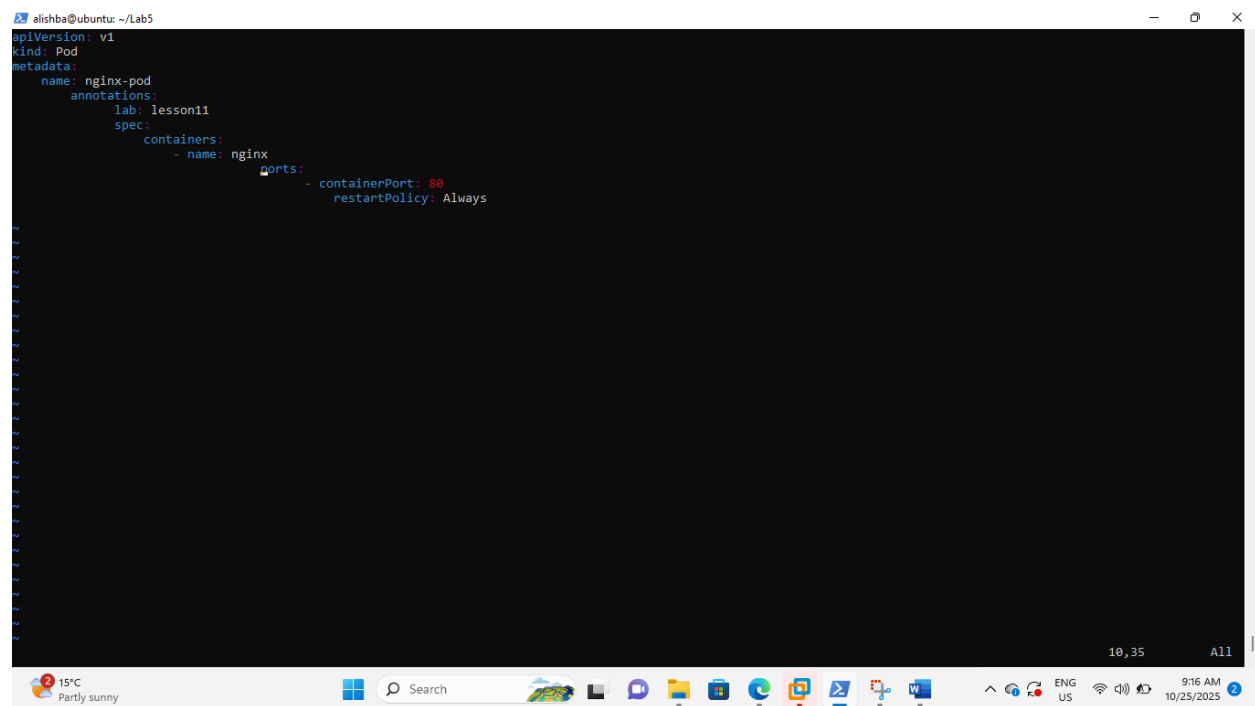
```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always
            # temp: do-not-keep
            #
```

```
restartPolicy: Always

alishba@ubuntu:~/Lab5$ vim k8s-sample.yaml
alishba@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

Task 11

Vim editing practice - delete, undo, numeric deletes, navigation



```
alishba@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        ports:
          - containerPort: 80
            restartPolicy: Always
```



```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
```

```
image: nginx:1.15
ports:
  - containerPort: 80
    restartPolicy: Always
```

Task 12

Vim search, add matches, substitute, undo

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - sample.yaml

      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```


$$\sim \frac{1}{r}$$

15°C Partly sunny Search ENG US 9:28 AM 10/25/2025


```
alishba@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  # webapp test
  # # webapp another
  #
  name: webapp-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - sample.yaml
        image: webapp:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

```
alishba@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  # nginx test
  # # nginx another
  #
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - sample.yaml
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```


Exam Evaluation Question:

Goal: This is an exam-style evaluation prompt. Students are asked to install Docker Desktop as part of the evaluation exercise on VM. No commands, solutions, hints, or step-by-step instructions are provided here — install Docker Desktop using your own knowledge and research.

```
alishba@ubuntu: /tmp

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
alishba@ubuntu: /tmp$ systemctl --user daemon-reload
alishba@ubuntu: /tmp$ systemctl --user enable --now docker-desktop
Created symlink /home/alishba/.config/systemd/user/graphical-session.target/wants/docker-desktop.service → /usr/lib/systemd/user/docker-desktop.service.
alishba@ubuntu: /tmp$ systemctl --user status docker-desktop --no-pager
● docker-desktop.service - Docker Desktop
   Loaded: loaded (/usr/lib/systemd/user/docker-desktop.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-10-25 05:20:43 UTC; 12s ago
     Main PID: 48739 (com.docker.back)
        Tasks: 74 (limit: 2207)
       Memory: 450.9M (peak: 451.4M)
          CPU: 8.887s
     CGroup: /user.slice/user-1000.slice/user@1000.service/app.slice/docker-desktop.service
             └─48739 /opt/docker-desktop/bin/com.docker.backend
                   └─48748 /opt/docker-desktop/bin/com.docker.backend services
                         └─48773 "/opt/docker-desktop/Docker Desktop" --name=error-dialog --socket=/home/alishba/.docker/desktop/error.reporter.sock
                               └─48776 "/opt/docker-desktop/Docker Desktop" --type=zygote --no-zygote-sandbox
                                     └─48777 "/opt/docker-desktop/Docker Desktop" --type=zygote
                                           └─48780 "/opt/docker-desktop/Docker Desktop" --type=zygote
                                                 └─48809 "/opt/docker-desktop/Docker Desktop" --type=zygote --no-zygote-sandbox
                                                       └─48817 /proc/self/exe --type=utility --utility-sub-type=network.mojom.NetworkService --lang=en-US --service-sandbox-type=none --enable-crash-reporter=a85...

Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.216116920Z][main.updater
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.218530836Z][main.component
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.218596774Z][main.apiproxy
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.219188373Z][main.ipc
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.219618714Z][main.updater
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.223749563Z][main.idle
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.229915977Z][main
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.230566777Z][main.report
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.232284685Z][main.report
Oct 25 05:20:45 ubuntu com.docker.backend[48739]: [05:20:45.232510832Z][main.ipc
Hint: Some lines were ellipsized, use -l to show in full.
alishba@ubuntu: /tmp$ docker --version
Docker version 28.5.1, build e180ab8
alishba@ubuntu: /tmp$
```

```
alishba@ubuntu: ~  
latest: Pulling from library/hello-world  
17ec7bbc9d7: Extracting [----->] 2.38kB/2.38kB  
docker: failed to register layer: exit status 22: unpigz: abort: zlib version less than 1.2.3  
  
Run 'docker run --help' for more information  
alishba@ubuntu:~$ sudo apt update  
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease  
Hit:3 https://download.docker.com/linux/ubuntu noble InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:5 http://archive.ubuntu.com/ubuntu jammy-security InRelease  
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease  
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
alishba@ubuntu:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages were automatically installed and are no longer required:  
  ca-certificates-java cpu-checker fonts-dejavu-extra ipxe-qemu ipxe-qemu-256k-compat-efi-roms java-common libaio1 libatk-wrapper-java libatk-wrapper-java-jni  
  libboost-iostreams1.74.0 libboost-thread1.74.0 libbrotli1.0.8 libcacard0 libdaxctl1 libdecor-0 libdecor-0-plugin-1-cairo libfdt1 libgfpapi0 libgfrpc0 libgfxdr0  
  libgif7 libglusterfs0 libiscsi7 libndctl6 libpmem1 libpmemobj1 librados2 librbd1 librdmacm1 libsd2-2.0-0 libspice-server1 liburing2 libusbredirparser1  
  libvirglrenderer1 msr-tools ovmf pass qemu-block-extra qemu-system-common qemu-system-data qemu-system-gui qemu-system-x86 qemu-utils qrencode seabios tree xclip  
Use 'sudo apt autoremove' to remove them.  
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:  
  libzvb1-common libavcodec60 libzvb10t64 libavutil58 libswresample4 7zip  
Learn more about Ubuntu Pro at https://ubuntu.com/pro  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
alishba@ubuntu:~$ sudo apt install --reinstall zlib1g zlib1g-dev -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Reinstallation of zlib1g is not possible, it cannot be downloaded.  
Some packages could not be installed. This may mean that you have  
requested an impossible situation or if you are using the unstable  
distribution that some required packages have not yet been created  
or been moved out of Incoming.  
The following information may help to resolve the situation:
```

```
alishba@ubuntu: ~/Downloads
Memory usage: 40%
Swap usage: 0%
IPv4 address for ens33: 192.168.145.131

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Oct 25 06:55:48 2025 from 192.168.145.1
alishba@ubuntu:~$ cd ~/Downloads
alishba@ubuntu:~/Downloads$ ls
docker-desktop-amd64.deb
alishba@ubuntu:~/Downloads$ sudo dpkg -i docker-desktop-amd64.deb
[sudo] password for alishba:
Selecting previously unselected package docker-desktop.
(Reading database ... 181217 files and directories currently installed.)
Preparing to unpack docker-desktop-amd64.deb ...
Unpacking docker-desktop (4.49.0-208700) ...
Setting up docker-desktop (4.49.0-208700) ...
Enabling use of privileged ports by Docker Desktop
kubernetes.docker.internal already in /etc/hosts
Reloading systemd daemon for logged in users
Done reloading systemd daemon for logged in users
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
alishba@ubuntu:~/Downloads$ sudo apt-get install -f -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

